# Mini Report - Exposed Rebar Inspection with YOLOv8

Detecting visible exposed reinforcement in concrete-inspection images to flag cases for human condition assessment

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## 1. Problem, use case and human role

**Problem.** Concrete-inspection photographs can contain visible exposed reinforcement, but manually reviewing many images is slow and may overlook cases that need attention.

**Use case.** A computer-vision model identifies visible exposed_rebar in inspection images and flags the image for review by a qualified engineer. The output is a bounding box and a segmentation mask with a confidence score; the mask is a screening aid, not a measurement.

**Human role.** The engineer reviews every flagged image and decides whether further condition assessment, testing or repair design is needed. A detection is an inspection flag, never a diagnosis; a missing detection does not prove the element is defect-free.

**Error cost.** A miss (false negative) is treated as roughly three times more costly than a false alarm: unflagged exposed reinforcement may never be inspected, whereas a false alarm only adds one human review.

## 2. Target class and labelling rules

| Item | Positive definition | Boundary cases | Box / mask rule |
|---|---|---|---|
| exposed_rebar | A reinforcing steel bar visibly exposed where the concrete cover is absent or broken. | Do not label cracks, including cracks parallel to reinforcement or with rust staining, unless a distinct visible steel bar can be seen. | One tight box per separate continuous visible rebar segment. Mask covers visible steel only; exclude surrounding concrete, voids, rust stains and shadows. Minimum target 20 × 20 px. |

## 3. Method and training configuration

Model: YOLOv8n-seg (Ultralytics) instance segmentation, trained on the Roboflow dataset rebar-exposure-qm020 v4 (1,547 training / 168 validation images). Training: 30 epochs, image size 640 × 640, batch size 8, random seed 42, NVIDIA Tesla T4 GPU, approximately 20 minutes. The selected checkpoint best.pt, the training curves (results.png), the epoch-level metrics (results.csv) and the Colab notebook are stored in the project repository, so the run can be repeated end to end.

## 4. Validation results and interpretation

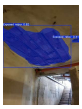
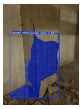
| Metric | Bounding boxes | Segmentation masks | Reading |
|---|---|---|---|
| Precision | 0.805 | 0.808 | About 1 in 5 flags is incorrect. |
| Recall | 0.703 | 0.717 | About 28 % of annotated instances are missed — the main risk. |
| mAP@50 | 0.772 | 0.762 | Good localisation at a loose overlap threshold. |
| mAP@50–95 | 0.529 | 0.470 | Mask outlines are coarse at strict overlap thresholds. |

Validation set: 168 images, 229 annotated instances. Mask metrics are the primary result, because the task requires spatial delineation and not only localisation. The gap between mAP@50 and mAP@50–95 is the quantitative counterpart of the qualitative review below: the model usually finds the right region but draws it too generously. Training curves show losses decreasing and metrics rising overall, with epoch-to-epoch fluctuation.

## 5. Error analysis on eight new inspection images

The trained checkpoint was applied to eight photographs outside the training workflow (images 1–4 and 6–9; image 5 was removed because it duplicated another photo in the set). Each prediction was reviewed visually against the labelling rules of Section 2.

| Image | Model outcome | Visual interpretation | Classification | Limitation | Action |
|---|---|---|---|---|---|
| **1**  | No detection | Continuous parallel bars near the upper edge, with apparent loss of cover. | **Suspected false negative (not yet confirmed)** | Low light, strong perspective, partial obstruction; small target. | Engineer review of the original photo; add low-light, wide-scene training images. |
| **2**  | 1 detection, 0.72 | Clear exposed mesh; several continuous bars visible. | **Likely true positive** | One broad mask covers several bars and surrounding concrete. | Flag for assessment; no area or quantity from the mask. |
| **3**  | 2 overlapping detections, 0.44 | Exposed bars lower in the soffit; predicted regions follow crack/spall geometry above. | **Suspected false positive; true bars missed** | Crack and spall geometry confused with steel; low confidence. | Human review; add crack/spall negatives and soffit-rebar examples. |
| **4**  | 4 detections, 0.84 / 0.66 / 0.42 / 0.30 | Spalled column face; no distinct continuous visible bar. | **Suspected false positives** | Spalling, dark cavities and texture read as rebar. | Human review; add damaged concrete without visible steel as negatives. |

| Image | Model outcome | Visual interpretation | Classification | Limitation | Action |
|---|---|---|---|---|---|
| **6**  | 2 detections, 0.79 / 0.74 | Rebar grid in the central spalled area; left region shows no clear steel. | **Central: likely true positive. Left: suspected false positive** | Large masks include surrounding damaged concrete. | Flag the central area; treat the left region as a false alarm. |
| **7**  | 2 detections, 0.82 / 0.41 | Stair/landing soffit; the main mask spans most of the underside. | **Likely true positive, heavy over-segmentation** | Mask follows the element, not the steel. | Close-range inspection; add tight-mask soffit images. |
| **8**  | 2 overlapping detections (0.68 + unscored) | Regular bar grid on a wall/column face; mask also covers spalled concrete and debris. | **Likely true positive** | Mask over-extends; duplicate boxes indicate weak NMS. | Flag for assessment; no steel area from the mask; tune NMS. |
| **9**  | 1 detection, 0.62 | Distant viaduct soffit near the pier head. | **Unverifiable at this resolution** | Visible steel cannot be confirmed at this scale. | Re-photograph at close range; set a minimum image scale. |

## 6. Data and model improvement actions

- Add negative examples of cracks, rust staining and spalled concrete without visible steel — the dominant false-positive sources (images 3, 4 and the left region of image 6).
- Add close-range soffit and overhead reinforcement images with tight masks, to correct the missed bars in image 3 and the over-segmentation in images 7 and 8.
- Add low-light, wide-scene and partially obstructed cases to recover misses of the type seen in image 1.
- Tune the confidence threshold and non-maximum suppression: duplicate regions (image 8) and low-confidence flags around 0.30–0.44 generate most of the review noise.
- Define an operational minimum image scale: distant views such as image 9 cannot be verified and should be re-photographed at close range.

## 7. Limitations and responsible use

The model cannot assess corrosion level, concrete strength, remaining bar diameter, structural capacity or repair urgency from an image alone. Performance depends on dataset coverage and is sensitive to lighting, shadows, occlusion, camera distance, viewpoint and surface texture. Masks systematically include adjacent damaged concrete, so neither the masks nor the slide illustrations may be used to compute damaged area, steel loss or repair quantities. Every prediction is an inspection flag for a qualified structural engineer, who remains responsible for any assessment, testing or intervention decision.

## 8. Conclusion

The prototype reaches 0.808 mask precision and 0.717 mask recall after 30 epochs and correctly flagged the clearest cases of exposed reinforcement (images 2 and 8, and the central region of image 6). Its weaknesses appear systematic rather than random: confusion between cracking or spalling and exposed steel, coarse mask boundaries, and misses in low-light or wide-scene photographs. Targeted data collection is the first improvement to test before a larger model, which makes the tool a credible first-pass screening aid for inspection photo sets — always under engineer review.